



Keeping Our Customers Successful





01

Company Profile



Initiative development (1983-2000)

Innovative development (2000-2010)

Transformation (2011-now)





Worldwide Services

500+
Engineers

Door to door global
engineering services.

5
Manufacturing Sites

China + Oversea manufacturing sites, **Shenzhen**,
Huizhou, Suzhou, Vietnam, Germany.

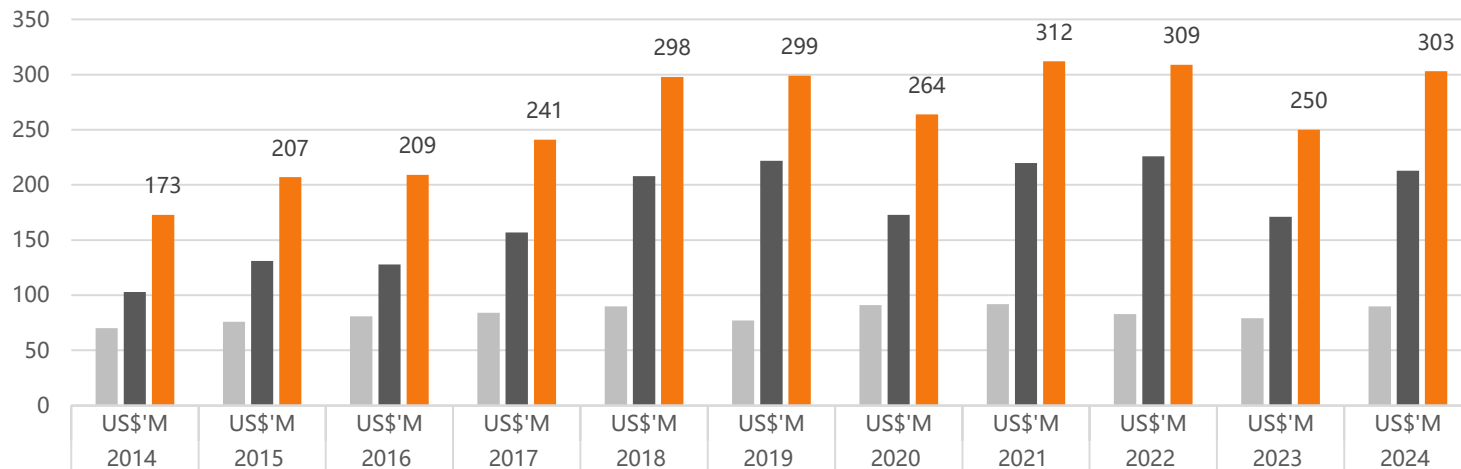
1
Mission

Make our customers
successful!

Sales Revenue

With on our leading technology and sincere services, we promote long-term and stable cooperation with our valued customers.

Ten-year Revenue — by business segments



■ Sales revenue - Molds	70	76	81	84	90	77	91	92	83	79	90
■ Sales revenue - Plastic parts	103	131	128	157	208	222	173	220	226	171	213
■ Total	173	207	209	241	298	299	264	312	309	250	303

■ Sales revenue - Molds

■ Sales revenue - Plastic parts

■ Total

Manufacturing Bases



China-Shen Zhen

- Tooling Fabrication
- Injection Molding
- Spray Painting
- Secondary Processes
- Mechanical Assembly
- Employees: 3260
- Floor space: 180,000m²



China-Hui Zhou

- Injection Molding
- LSR Molding
- Spray Painting
- Secondary Processes
- Electro-Mechanical Assembly
- Employees: 900
- Floor space: 32,000m²



China-Su Zhou

- Tool Fabrication
- Injection Molding
- Spray Painting
- Secondary Processes
- Electro-Mechanical Assembly
- Employees: 450
- Floor space: 10,000m²



Vietnam

- Basic Secondary Processes
- Injection Molding Machines: 38
- Employees: 200
- Floor Space: 20,000m²
- Basic Secondary Processes
- ECN + Tooling Modification capability



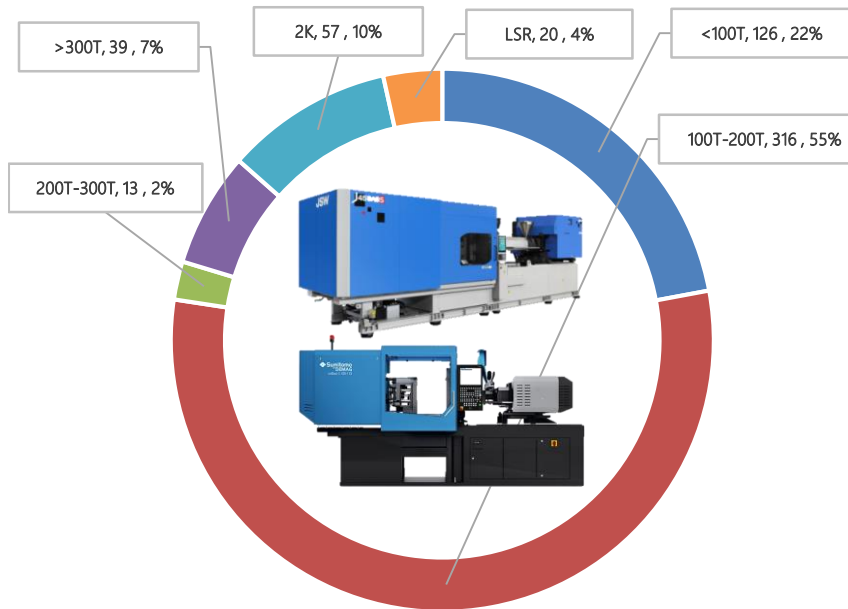
02

Capacity & Capabilities & Technologies



Injection Molding Machines

- Total 600+ Injection Machines
- High Speed & Electrical Machines
- Machine Range 6T~550T
- 60+ Double Shot Machines
- Multi-shots by AUI platform
- Triple Shot Machines
- LSR Injection Machines
- Vertical Injection Machines



Injection machine	Brand	Ton	Country	Total
<200T	JSW	55T	Japan	2
		75T	Japan	1
		85T	Japan	57
		100T	Japan	14
		110T	Japan	80
		140T	Japan	7
		150T	Japan	7
		180T	Japan	90
	Sumitomo	180T	Japan	80
	Fanuc	50T	Japan	1
		100T	Japan	8
		150T	Japan	19
	TOYO	100T	Japan	20
		130T	Japan	2
		180T	Japan	1
	Babypplast	6T	Italy	1
	Sodick	100T	Japan	10
	Hai Tian	160T	China	5
LSR	Arburg	60T	Germany	10
	KraussMaffei	80T	Germany	5
		100T	Germany	5
>200T	JSW	220T	Japan	3
		280T	Japan	4
		350T	Japan	14
		450T	Japan	18
		450T	Japan	1
	Fanuc	300T	Japan	2
	TOYO	280T	Japan	4
		450T	Japan	1
Double Shot (2K)	KraussMaffei	200T	Germany	2
	JSW	180T	Japan	40
	Nissei	140T	Japan	4
		200T	Japan	1
	FCS	140T	Taiwan	1
		260T	Taiwan	6
	TOYO	230T	Japan	2
	JSW	350T	Japan	3
High speed	JSW	550T	Japan	1
	Sodick	100T	Japan	8
	Fanuc	100T	Japan	2
		130T	Japan	5
Vertical Machine	Sumitomo	180T	Japan	10
		85T	Taiwan	12

High-End Precision Machines

- High precision machines from Europe/Japan/US
- Machine Brand - Röders / Yasda / Makino / Agie Charmilles / Hauser / Studer
- Machining tolerance up to 0.002mm

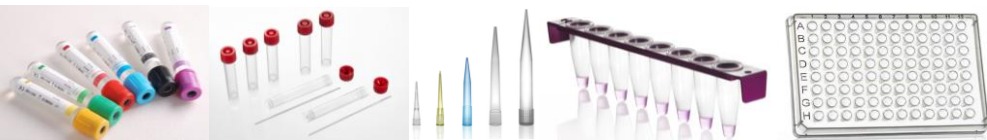


Major Machines	Quantity
CNC Milling Machine	104
5-Axis CNC Milling Machine	12
EDM Machine	55
EDM Wire-cut Machine	19
Grinding Machine	59
CNC Jig grinder	2
EDM Hole Drilling Machine	3
Gun Drilling Center	4
Lathe	6
Mold Reversal Machine	10
High Precision Hard Turning	1
CMM	17
IMM for test (Including 2k)	18(6)



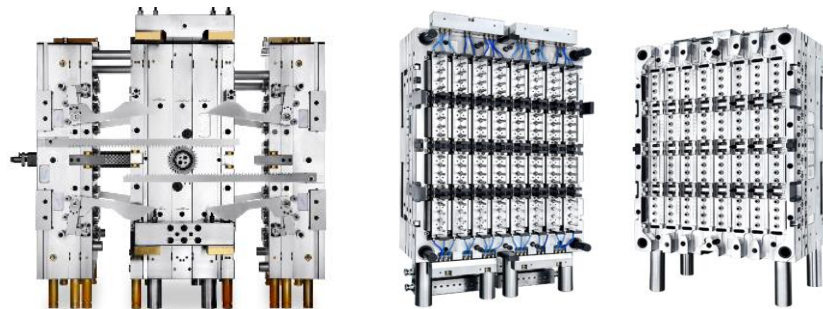
Full Automation Line

- 4 Full Automation Lines;
- Integrated with CNC and EDM machining processes, CMM is build in the line for inspection, line length ~ 51m;
- 7 days / 24 hours production with MES system monitoring;



Multi-Cavitation Mold & Stack Mold

- Experiences in Multi-cavity and stack molds design, manufacturing and qualification;
- Exchangeable inserts by high precise machining
- Cavity number from 1~256;
- In-house tool trial machines 50T~1500T



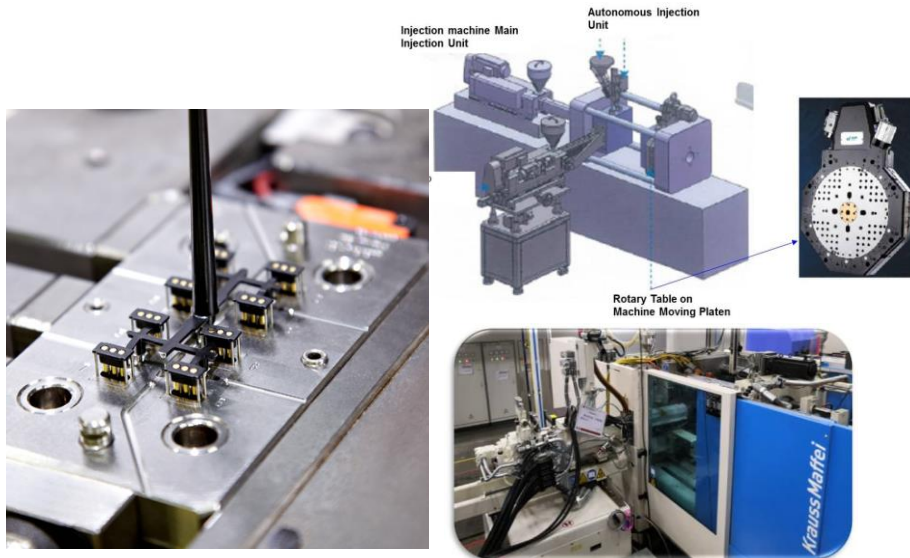
Medical Mold & Packaging Mold

- TK has rich experiences serving top tier medical customer, PCR, POCT and blood collection components;
- Decades of experience in packaging molds, with the ability from packaging products design, design certifications, mold development and full automation production capabilities.



Insert Mold & Over Mold & Multi-Shots Mold

- 2K & 3K (Double shot & Triple shot) molds design, manufacturing and qualification;
- Parallel barrel double shot & "L" type double shot;
- Multi-shots solutions by AUI (Autonomous injection unit);
- Precise insert mold, waterproof up to 7 bar (equal to IP68);

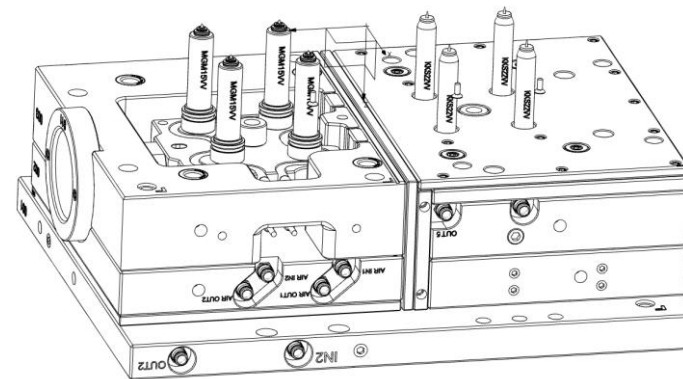


LSR Mold

- Single shot LSR mold & LSR over mold & Double shot (2K) LSR mold;
- Flash free by unique sealing system >99.8% vacuum in mold + up to 0.01mm parting line;
- Rapid prototyping solution by TK standard mold base and cold runner system, give Lower Cost & Short Lead Time, 2 weeks prototype service for single LSR.

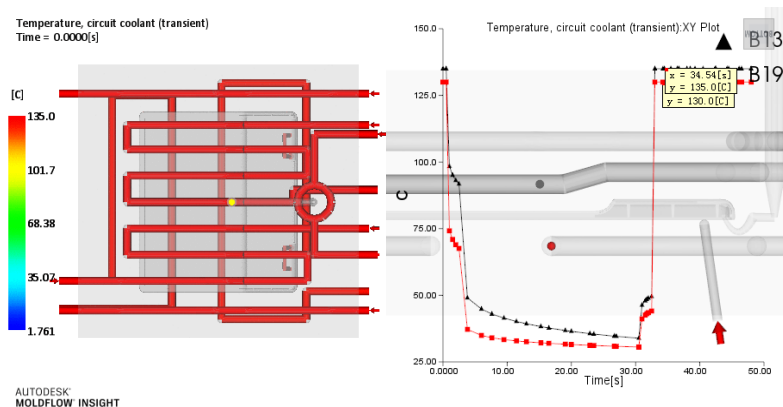
Hot runner system for engineering plastic

Cold runner system for LSR



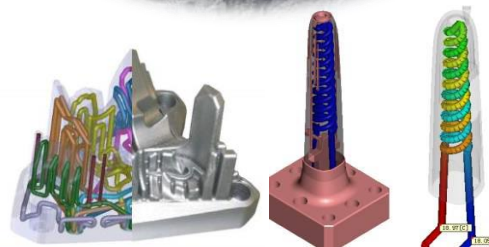
Rapid Heating Cycle Molding (RHCM)

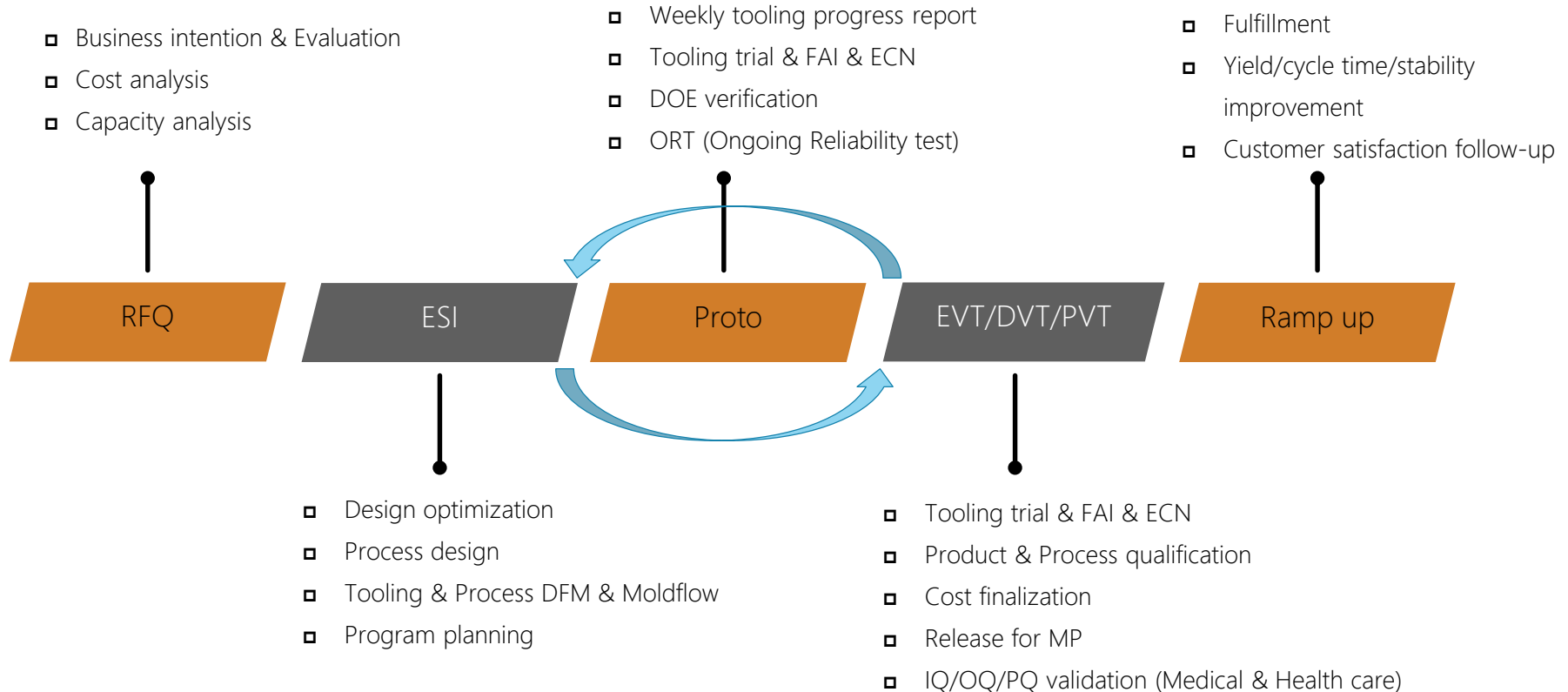
- Experiences using Electric Heating / High-pressure Steam Heating / Conductive Heating to achieve RHCM
- Experience working with RocTool / Single
- Advanced technology to achieve thin wall, high gloss, or reduce stress mark and welding line, etc.



Conformal Cooling

- In-house DMLS technology (Direct Metal Laser Sintering);
- Diffusion Welding Technology;
- Cooling efficiency improve up to 40%;
- Reducing cosmetic defects.





Molding Capability



IML/IMD

- Over 10-year experience in precision IML/IMD manufacturing;
- In-house one-stop IML services;
- Offset printer / digital printer / silk screen printing;
- Thermal forming / high pressure forming.
- Dead front / Personalized graphic / Waterproof window / conductive circuit, etc.



2K/3K

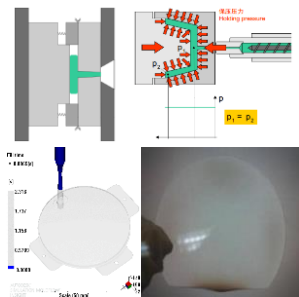
Double/Triple shots

- 60+ 2K machines;
- Multi-shot with UAI solution;
- 1st shot / 2nd shot / 3rd shot raw material selection experience;
- Multi-shots interlock feature design service;



LSR Molding

- Liquid silicone injection molding;
- Liquid compression silicone molding;
- LSR over molding with engineering plastic/Metal/Silicone
- Double shot 2K LSR + LSR or LSR + Engineering plastic or LSR + Glass;
- Fast turn-around prototype service with standard mold base.



Injection Compression Molding

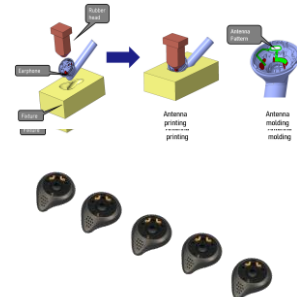
ICM

- Inner stress elimination;
- Reduce rainbow pattern in Lens products;
- Help with thin wall product molding production;



Precise Insert Molding

- Vertical injection machines;
- Metal insert molding;
- Low pressure insert molding for PCB board;
- Nano molding;
- Glass insert molding
- Fully automatic production;
- For waterproof product, up to 7 bar air leakage control;



PDS

- Conduction Antenna Technology
- Direct printing to reduce manufacturing costs with plastic/metal/glass
- High Reliability, Low Cost, Environmental Friendly than of LDS

Secondary Process Capability

- Printing
 - ✓ Pad Printing
 - ✓ Silk screen printing
 - ✓ 6 color offset printing
 - ✓ Digital printing
- Spray Painting
 - ✓ 4 spindle lines
 - ✓ 450m~560m length
 - ✓ 3 Painting & 3 Roasting Station
 - ✓ NCVM
 - ✓ Class 100K Clean Room
 - ✓ Class 10K Clean Spray Booth
- Lamination
- CNC Shaping & De-gating
- Laser Etching
- Ultrasonic Welding
- Solvent Painting
- Waterborne Painting
- NCVM
- Foil Stamping
- Fabric Warping
- Laser Direct Structuring (LDS)
- Printed Direct Structuring (PDS)
- Polishing



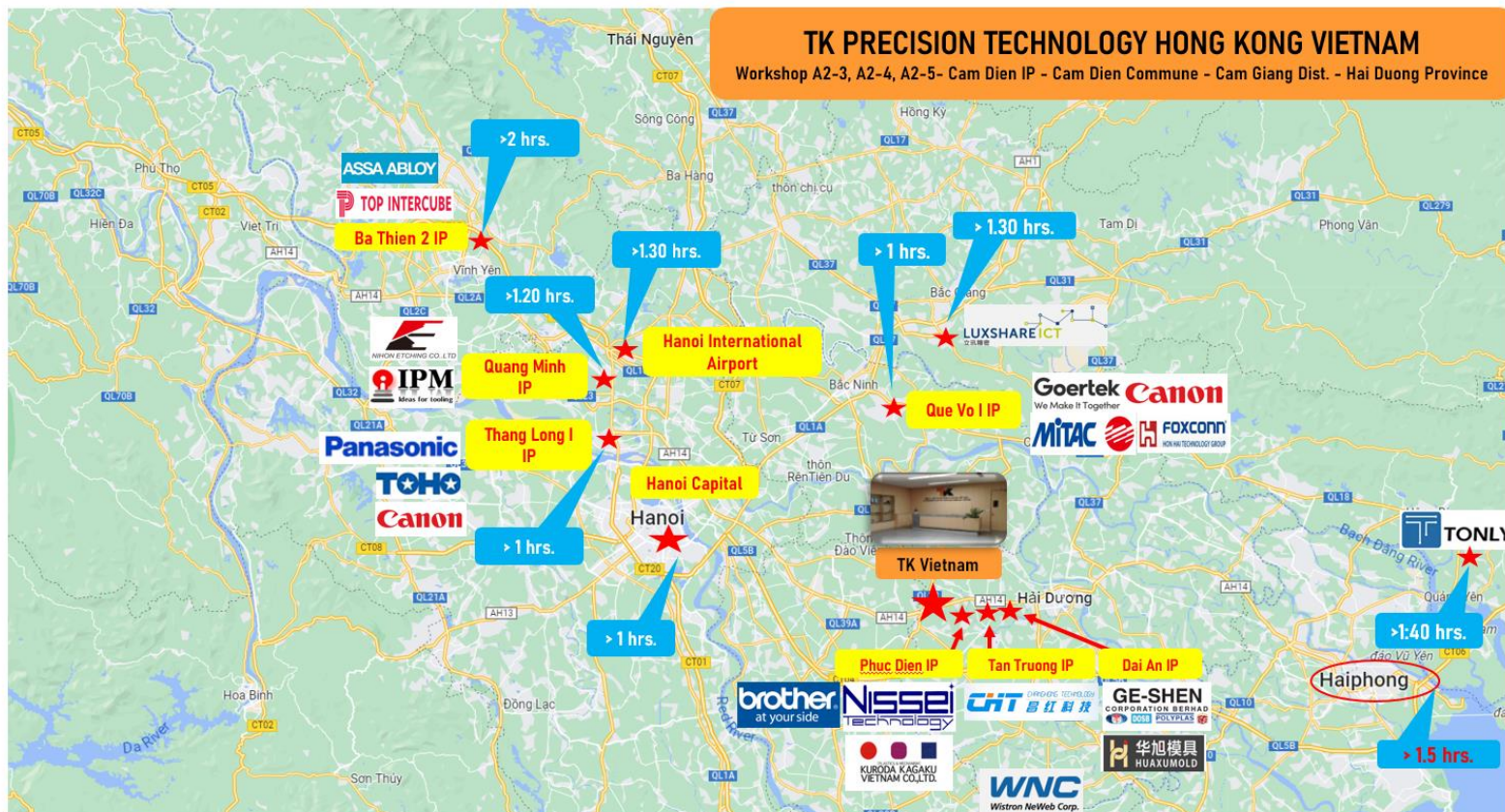


03

TK Vietnam Introduction



TK Vietnam – Geographic Location – 1.5 hours to HaiPhong Harbor



ISO 9001



ISO 14001



ISO 45001



UL Certification



TK Vietnam – General Information

TK Vietnam factory was established in Q1, 2021 at Hai Duong province, VSIP (Vietnam Singapore Industrial Park).

This factory was operational ready since Q1, 2022, same mission to provide Global Leading one-stop plastics solution. dedicated to the development and manufacturing of plastic molds and parts in mobile phone, healthcare, smart home, packaging, medical and consumer electronic etc.



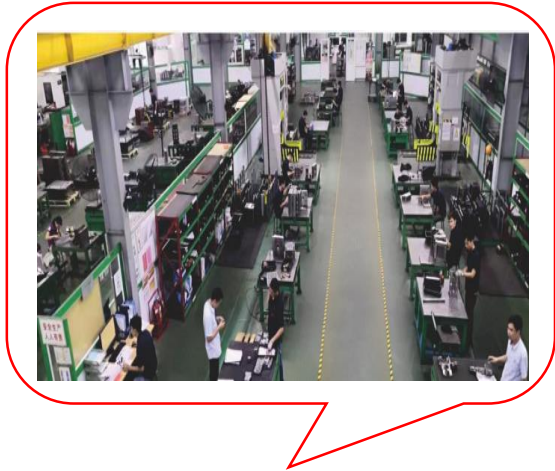


	Equipment	Brand	Size	QTY	Current Status
Production	Injection Machine Factory -Total 28 IM	JSW	85T	1	Single shot(Ready)
		JSW	100T	9	Single shot(Ready)
		JSW	180T	10	Single shot(Ready)
		Sumitomo	350T	4	Single shot (Ready)
		Haitian	160T	1	Horizontal 2K(Ready)
		Haitian-Zhafir	230T	2	Side-mount 2K(Ready)
	Injection Machine NPI- Shenzhen -Total 6 IM	Haitian-Zhafir	300T	1	Side-mount 2K (Ready)
		Sumitomo	180T	2	Single shot(Ready)
		Engel	220T	1	2K (Ready)
		Haitian	300T	1	2K (Ready)
		JSW	350T	1	Single shot(Ready)
		Sumitomo	350T	1	Single shot(Ready)

- ✓ Glue dispensing
- ✓ CNC
- ✓ Pad & Silk screening printing
- ✓ Foil stamping
- ✓ Heat staking
- ✓ Laser Etching
- ✓ Ultrasonic Welding

- Expanded to 2 molding work shop in 2024
- Keep adding machine in 2025&2026
- Enlarge warehouse and secondary process (IML, CNC, printing etc.)

	Equipment	Brand	Q'ty	Current Status
Tool Repair & Maintenance	Surface Grinder	Okamoto	1	Ready
	Milling Machine	Okamoto	1	Ready
	CNC Machining	Makino	1	Ready
	EDM Machine	GF	1	Ready



TKVN Tooling Outsource Partner

- Tool repair and spare parts
- Simple tool fabrication



科时唯一（越南）有限公司
FORCE UNIQUE VIET NAM CO.,LTD



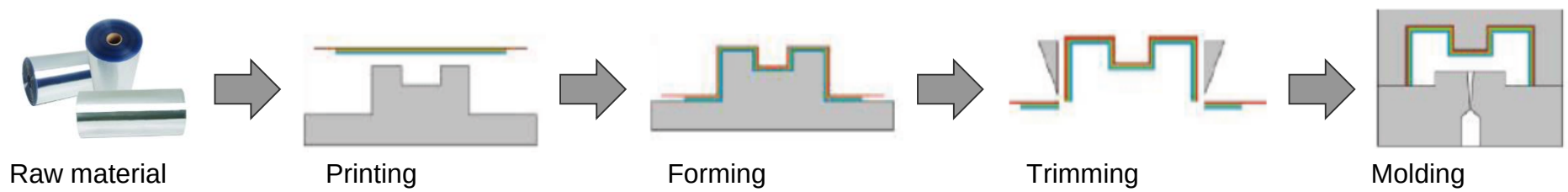
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IML Introduction



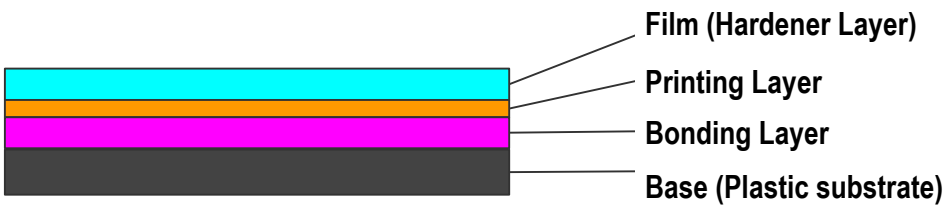
IML Capability (In Mold Label)

Process:



Features:

- Good wear-resistance and colorful
- Capable to replace the painting, plating, VM etc. process



Printing:

- 6 color Offset printing
- Digital Printing
- Silk Screen Printing
- Pad Printing



Forming:

- Thermal Forming
- Vacuum Forming
- High Pressure Forming

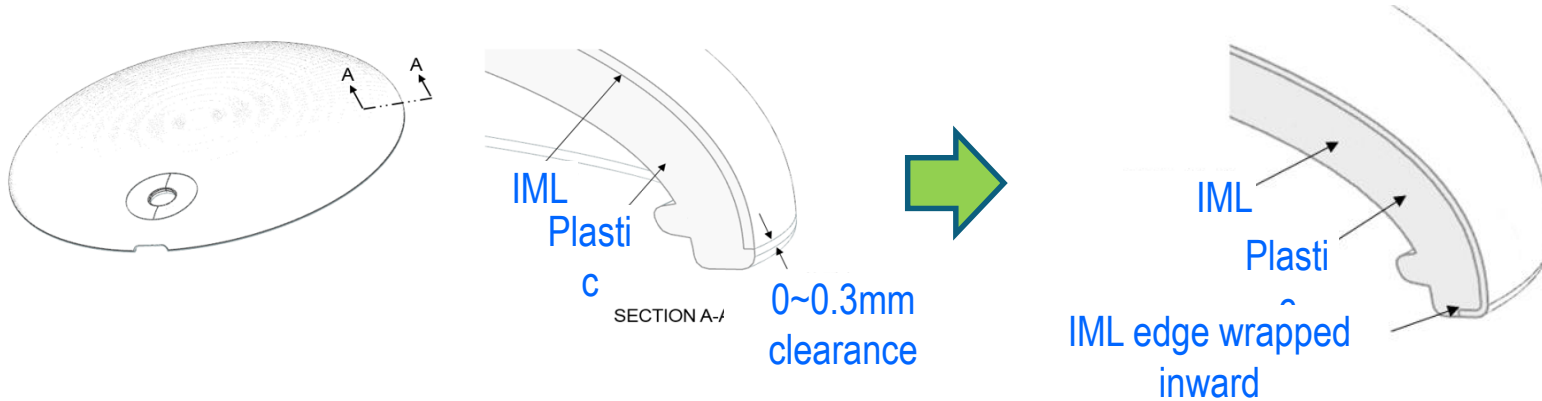


IML Capability

Parting Line Wrapped IML – Case Study

Purpose:

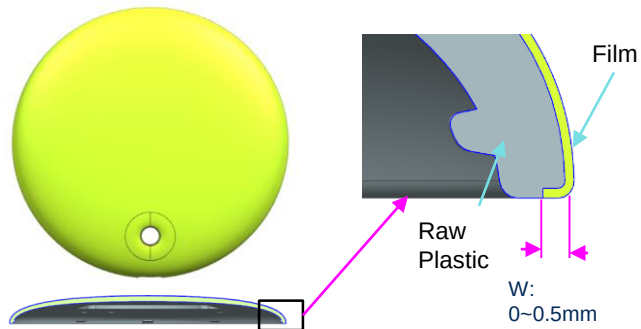
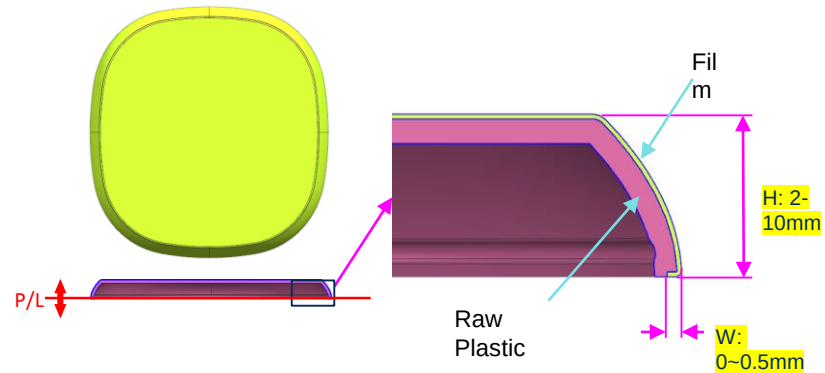
In general IML application, there is 0~0.3mm clearance between IML edge and P/L. This clearance is always complained by PD / ID.



IML Capability

Parting Line Wrapped IML

- Dead-Front appearance
- Stop light leakage at P/L
- Clean Parting Line

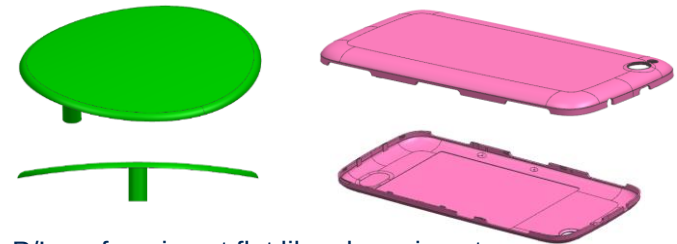


PS.

Parting line surface must be flat.

Film option: PC, PC+PMMA

Film thickness: 0.175mm, 0.25mm



P/L surface is not flat like above is not suitable for wrapped IML.

IML Capability

Parting Line Wrapped IML

- The shape is not limited
- (can be round, ellipse, rectangle and any other shape);



Switch Off
Not Visible

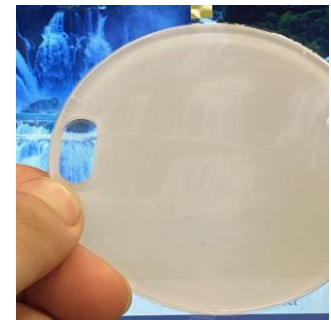


Switch On
Visible Light

Polarized IML



Not Visible
No Light Pass Through
from Background



Visible after Rotate 90°
Light Pass Through

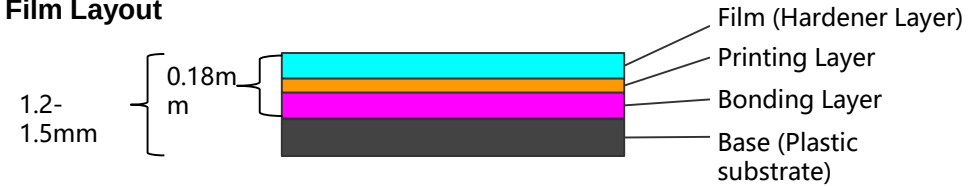
IML Capability

IML Solution Replace PVD

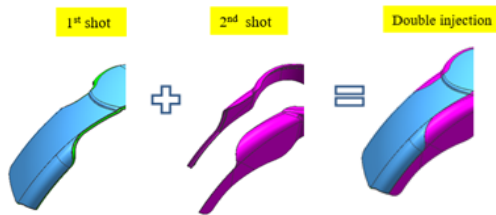
Features for IML Solution:

- Good Appearance, Colorful, Metallic glossy
- Good Wear-Resistance and Reliability Performance
- Reduce Cost

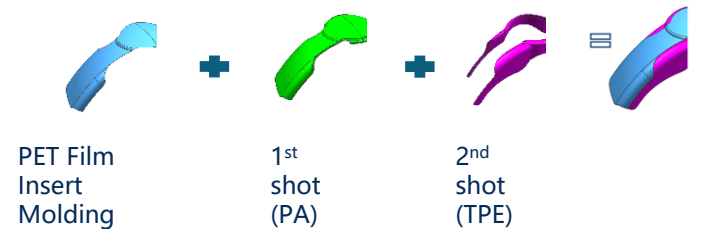
Film Layout



Current Process, Yield ~50%
2k Injection > PVD 2nd Process



New Solution, Yield Rate Improved, >70%
IML Process 2k with IML Solution



IML Capability

Functional IML:

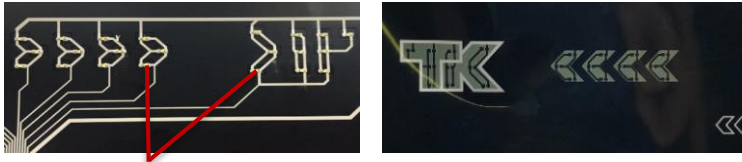
- Feature on both surfaces.
- Our knowhow: Film & ink selection / Process design / Protect circuit during forming
- Application: Electronic Products, Home Appliances , Automotive, etc.



Front side of film can print with graphics.



Back side can print with electronic circuits.



SMT components



Antenna or RF circuit integrate on IML

thank you!

